# Smart Medi Box - Wiring Guide

Professional Edition

## System Overview

The Smart Medi Box runs from a 12V supply split into two rails for power isolation.

### 12V Rail (Direct)

- Solenoid Lock
- Optional Cooling System

### 5V-6V Rail (via buck converter)

- Servo Motor

### 5V Rail (via regulator)

- Arduino Leonardo
- SIM800L EVB
- Sensors (DHT22, DS18B20, RTC)
- LCD Display
- RFID Module (via 3.3V pin)

## Power System Design

### Input Supply

`12V / 5A DC Power Supply (recommended)`

### Voltage Regulation

**Recommended:** Buck Converter (LM2596)
**Alternative:** LM7805 (with heatsink)

### Capacitors (Essential)

- Input: 100uF electrolytic
- Output: 10uF electrolytic + 100nF ceramic
- SIM800L: 1000uF near VCC

> Linear regulators generate heat: Power Loss = (12V - 5V) x Current. Use a buck converter for efficiency.

Critical rule: All components must share a common ground (GND).

## Microcontroller (Arduino Leonardo)

### Power

- VCC -> 5V
- GND -> Common Ground

## Pin Configuration

### Digital Pins

| Pin | Function |
|---|---|
| 0 | SIM800L RX |
| 1 | SIM800L TX |
| 2 | Door Switch |
| 3 | Cooling (MOSFET Gate) |
| 4 | Stepper STEP |
| 5 | Solenoid (MOSFET Gate) |
| 6 | RFID RST |
| 7 | Buzzer |
| 8 | LCD Reset |
| 9 | RFID CS |
| 10 | LCD RS |
| 11 | LCD RW |
| 12 | Stepper DIR |

| Pin | Function |
|---|---|
| 13 | LCD E |

### Analog Pins

| Pin | Function |
|---|---|
| A0 | DHT22 Data |
| A1 | DS18B20 Data |
| A2 | Servo Signal (MG995) |

## Hardware Compatibility Check

- SIM800L EVB: Requires stable 5V with high peak current. Keep 1000uF cap close to VCC/GND.
- RC522 RFID: Must use 3.3V logic. Do not connect VCC to 5V. Keep SPI lines short.
- ST7920 LCD: Uses ST7920 serial (not standard SPI). Verify PSB to GND for serial mode.
- DHT22 + DS18B20: Use 4.7K pull-up on data. Keep data lines away from motor wiring.
- Servo MG995: Use 5V-6V only. Do not power from 12V rail.
- Solenoid: Must use flyback diode across coil. MOSFET required.
- Shared GND: All modules must share a common ground reference.
- Door switch on D2 conflicts with I2C SDA. If RTC is used, move the door switch to a free pin.

## Power Wiring Diagram (ASCII)

```
12V DC IN
   |
   +--------------------> Servo (5V-6V supply via buck)
   |
   +--------------------> Solenoid (12V) -> MOSFET -> GND
   |
   +--> Buck Converter (LM2596) -> 5V RAIL
           |
           +--> Arduino Leonardo 5V
           +--> SIM800L EVB 5V (1000uF cap near VCC/GND)
           +--> LCD 5V
```

```
            +--> Sensors 5V
            +--> Stepper Driver (5V logic + separate motor supply 6V-12V)
            +--> RC522 3.3V (from Arduino 3.3V pin)

ALL GROUNDS TIED TO COMMON GND
```

## Arduino Leonardo Full Pinout (Reference)

### Digital Pins

| Pin | Function | Notes |
|---|---|---|
| 0 | RX | UART RX (Serial1) |
| 1 | TX | UART TX (Serial1) |
| 2 | SDA | I2C SDA |
| 3 | SCL | I2C SCL |
| 4 | GPIO | Digital I/O |
| 5 | PWM | Digital I/O, PWM |
| 6 | PWM | Digital I/O, PWM |
| 7 | GPIO | Digital I/O |
| 8 | GPIO | Digital I/O |
| 9 | PWM | Digital I/O, PWM |
| 10 | PWM | Digital I/O, PWM |
| 11 | PWM | Digital I/O, PWM |
| 12 | GPIO | Digital I/O |
| 13 | LED | Digital I/O, onboard LED |

### Analog Pins

| Pin | Function | Notes |
|---|---|---|
| A0 | Analog In | ADC0 |
| A1 | Analog In | ADC1 |
| A2 | Analog In | ADC2 |
| A3 | Analog In | ADC3 |
| A4 | Analog In | ADC4 |
| A5 | Analog In | ADC5 |

### Power Pins

| Pin | Function |
|---|---|
| VIN | External input (7V-12V) |
| 5V | Regulated 5V output |
| 3.3V | 3.3V output (for RC522) |
| GND | Ground |
| RESET | Reset input |

### SPI Pins

- **IMPORTANT (Arduino Leonardo):** Use ICSP header for SPI (MOSI, MISO, SCK).
- D11-D13 are not used for SPI on Leonardo. Use ICSP for RC522.

### USB Serial

- USB serial is separate from pins 0/1 on Leonardo.

## SIM800L EVB (GSM Module)

### Connections (4 wires only)

| Pin | Connection |
|---|---|
| VCC | 5V |
| GND | GND |
| TX | Arduino Pin 0 (RX) |
| RX | Arduino Pin 1 (TX) |

Note: TX/RX must be crossed (TX -> RX, RX -> TX).

### Requirements

- SIM card (correct orientation)
- Antenna connected
- 1000uF capacitor for stability

Important: SIM800L draws up to 2A peak current. Ensure:
- Thick power wires
- Short connection to power source
- 1000uF-2200uF capacitor near module
- Do NOT power from Arduino 5V pin

## LCD Display (12864B V2.3 - ST7920)

### LCD Type & Mode

- **Model:** ST7920 128×64
- **Mode:** Serial (SPI-like mode) when PSB = GND
- ⚠️ **IMPORTANT:** PSB must be GND for serial mode. If PSB = 5V, it switches to parallel mode and U8g2 wiring/constructor must change.

### Connections

| LCD Pin | Function | Arduino Pin |
|---|---|---|
| PSB | Mode Select | GND (Serial mode) |
| VCC | Power | 5V |

| LCD Pin | Function | Arduino Pin |
|---|---|---|
| GND | Ground | GND |
| RS | Register Select | Pin 10 |
| RW | Read/Write | Pin 11 |
| E | Enable | Pin 13 |
| RST | Reset | Pin 8 |

**Library:** U8g2

### U8g2 Constructor (Working - Serial/SPI Mode)

```
#include <U8g2lib.h>
U8G2_ST7920_128X64_F_SW_SPI u8g2(U8G2_R0, 13, 11, 10, 8);
// Parameters: (rotation, clock=Pin13(E), data=Pin11(RW), cs=Pin10(RS), reset=Pin8(RST))
```

Uses ST7920 serial protocol (not standard SPI). PSB = GND → Serial mode (this config). PSB = 5V → Parallel mode (requires different constructor & wiring).

### Backlight

- 🔆 **BLA** (Backlight Anode, +) → 5V
- ⚫ **BLK** (Backlight Cathode, −) → GND

## Sensors

### DHT22 (Temperature & Humidity)

**Required Libraries:** DHT sensor library, OneWire, DallasTemperature

**Wiring:**

- VCC → 5V
- GND → GND
- DATA → A0
- **4.7K pull-up resistor required** between DATA and 5V (or 3.3V)

### DS18B20 (Temperature)

**Wiring:**

- VCC → 5V
- GND → GND
- DATA → A1
- **4.7K pull-up resistor required** between DATA and 5V (or 3.3V)

### RTC DS3231 (I2C)

**Required Library:** RTClib by Adafruit

**DS3231 Wiring on Leonardo:**

- Use the **SDA/SCL labeled pins** (near AREF), NOT D2/D3
- DS3231 SDA → SDA header pin
- DS3231 SCL → SCL header pin
- VCC → 5V
- GND → GND

> ⚠️ Important: On Leonardo, use the dedicated SDA/SCL header pins (near AREF), not the D2/D3 digital pins. This avoids conflicts with the door switch on D2.

### RTC Time Set (Compile Time)

```
#include <Wire.h>
#include <RTClib.h>

RTC_DS3231 rtc;

void setup() {
  Serial.begin(9600);
  delay(1000);

  if (!rtc.begin()) {
    Serial.println("Couldn't find DS3231 (check SDA/SCL pins).");
    while (1) {}
  }

  Serial.println("RTC will be set to COMPILE TIME after countdown.");
  for (int i = 10; i >= 1; i--) {
    Serial.print("Setting in ");
    Serial.println(i);
    delay(1000);
  }
```

```
  rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));
  Serial.println("Done. RTC set.");
}

void loop() {}
```

## RFID Module (RC522)

### Connections (ICSP Header)

**RC522 Pin → Arduino Pin/Header:**

- RC522 MOSI → ICSP MOSI
- RC522 MISO → ICSP MISO
- RC522 SCK → ICSP SCK
- RC522 VCC → **3.3V** (NOT 5V)
- RC522 GND → GND
- RC522 SDA/SS (CS) → Pin 9
- RC522 RST → Pin 6

> ⚠️ **CRITICAL NOTES:**
> ✓ RC522 operates at 3.3V logic level. Do NOT connect VCC to 5V.
> ✓ Use ICSP header for SPI (not digital pins D11-D13 on Leonardo).
> ✓ ICSP pinout diagram is attached in the build pack.
> ✓ Consider using a logic-level converter or resistor dividers for MOSI, SCK, and CS lines if power supply is not stable.

## Motors and Actuators

### Stepper Motor (EasyDriver HW-135)

- STEP -> Pin 4
- DIR -> Pin 12
- VCC -> 5V (logic)
- M+ -> 6V-12V (motor supply)
- GND -> GND

### Servo Motor (MG995)

- Signal → A2 (digital pin 20)
- VCC → 5V–6V (NOT 12V)
- GND → GND

> On Leonardo, analog pins can be used as digital I/O. Using `s.attach(A2)` is valid (A2 is also known as digital pin 20). Use external 5V supply if needed.

### Solenoid Lock (12V)

- Positive -> 12V
- Negative -> MOSFET Drain

### MOSFET (N-Channel)

- Gate -> Pin 5 (via 1K resistor)
- Drain -> Solenoid
- Source -> GND

### Protection

- Diode (1N4007) across solenoid

### Buzzer (TMB12A05 + BC337 Driver)

**Buzzer Type:** Active electromagnetic buzzer

**Wiring:**

- Arduino D7 → 1K resistor → BC337 BASE
- BC337 EMITTER → GND
- BC337 COLLECTOR → Buzzer (-)
- Buzzer (+) → +5V

**Protection Diode (Recommended):**

- Use 1N4148 or 1N4007 across the buzzer coil
- Diode cathode (stripe) → +5V (buzzer +)
- Diode anode → Buzzer (-) / BC337 collector

**Common Ground:** Arduino GND must be connected to the buzzer supply GND

**Operation:** This is an ACTIVE buzzer—it makes sound with DC voltage

- Use `digitalWrite(D7, HIGH)` to turn ON
- Use `digitalWrite(D7, LOW)` to turn OFF

### Cooling System (Pin D3) – MOSFET Low-Side Switch

**Supported Loads:**

- DC fan (2-wire)
- Peltier / TEC module (thermoelectric cooler)
- Any DC load within MOSFET + power supply limits

**Wiring (LOW-SIDE MOSFET SWITCH):**

- Arduino D3 → 1K resistor → MOSFET GATE
- MOSFET SOURCE → GND (power supply negative)
- MOSFET DRAIN → Load (-) (fan- or TEC-)
- Load (+) → +V supply (12V typical for fan/TEC)

**CRITICAL REQUIREMENTS:**

- **1) Common Ground Required:** Arduino GND must be connected to the load power supply GND
- **2) Gate Pulldown (Recommended):** Add 100K (or 10K) resistor from MOSFET GATE to GND to prevent floating gate
- **3) Flyback Diode:** REQUIRED for inductive loads (DC fan, motor, relay). NOT used for TEC/Peltier.
- **4) Peltier (TEC) WARNING:** TEC modules draw HIGH current (3A–10A+). Use a separate PSU and logic-level MOSFET. Consider a fuse.
- **5) PWM Control:** D3 supports PWM. For TEC, PWM can reduce efficiency; consider a thermostat/PID approach for stable temperature.

### Door Switch (Pin D2)

**Pin Configuration:** Pin 2 with INPUT_PULLUP

```
const int DOOR_PIN = 2;

void setup() {
  Serial.begin(9600);
```

```
  pinMode(DOOR_PIN, INPUT_PULLUP);
}

void loop() {
  int state = digitalRead(DOOR_PIN);
  if (state == LOW) {
    Serial.println("Door CLOSED (switch closed to GND)");
  } else {
    Serial.println("Door OPEN (switch open)");
  }
  delay(100);
}
```

## Power Budget

### 5V Rail

Approx: 350mA typical | Peak: ~600mA

### 12V Rail

Servo + Solenoid: up to ~1.8A

### SIM800L

Peak bursts: ~2A

Power supply: 12V / 5A recommended (SIM800L + servo peaks)

## Protection (Recommended)

- Add inline fuse (2A-5A) on 12V input
- Add reverse polarity protection diode

## Key Design Rules

- All grounds must be connected together
- Use 3.3V for RFID (RC522)
- Servo must use 5V-6V only (not 12V)

- Add capacitors for stability
- Avoid pin conflicts
- Use short wires for communication lines
- Linear regulators generate excessive heat (use buck converter)

## First Power-On Procedure

1. Verify 5V and 12V using multimeter
2. Power Arduino only -> test code
3. Connect SIM800L -> test AT command
4. Add sensors one by one
5. Test actuators last

## Troubleshooting Essentials

### No GSM Response

- Check TX/RX wiring
- Check power stability
- Verify baud rate (115200)

### Sensor Failure

- Check pull-up resistors
- Verify pin connections

### Motor Issues

- Check driver wiring
- Verify control signals

### System Resets

- Add bulk capacitor (1000uF)
- Improve grounding

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